

**Advanced Econometrics – Exam**

June 20, 2022

**Exercise 1 (30%)**

In this example, we have (synthetic) data<sup>1</sup> on whether workers complain to managers at fast-food restaurants. The covariates are **age** (in years of the worker), **grade** (years of schooling completed by the worker), **south** (equal to 1 if the restaurant is located in the South), **tenure** (the number of years spent on the job by the worker), **gender** (of the worker, 1 for women), **race** (of the worker), **income** (in thousands of dollars by the restaurant), **genderm** (gender of the manager, 1 for women), **burger** (equal to 1 if the restaurant specializes in hamburgers), and **chicken** (equal to 1 if the restaurant specializes in chicken).<sup>2</sup>

Call:

```
glm(formula = complain ~ age + grade + south + tenure + gender +
     income, family = binomial(link = "probit"), data = chicken)
```

Coefficients:

|             | Estimate   | Std. Error | z value | Pr(> z )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 0.4023853  | 0.5083557  | 0.792   | 0.42863      |
| age         | -0.0336174 | 0.0118943  | -2.826  | 0.00471 **   |
| grade       | 0.0331282  | 0.0243458  | 1.361   | 0.17360      |
| south       | 0.0776746  | 0.0547301  | 1.419   | 0.15583      |
| tenure      | -0.0029834 | 0.0899963  | -0.033  | 0.97355      |
| gender      | 0.2975866  | 0.0547633  | 5.434   | 5.51e-08 *** |
| income      | -0.0020855 | 0.0006771  | -3.080  | 0.00207 **   |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2804.7 on 2762 degrees of freedom

Residual deviance: 2752.6 on 2756 degrees of freedom

AIC: 2766.6

Call:

```
probitmfx(formula = complain ~ age + grade + south + tenure +
           gender + income, data = chicken, atmean = TRUE)
```

Marginal Effects:

|        | dF/dx       | Std. Err.  | z       | P> z          |
|--------|-------------|------------|---------|---------------|
| age    | -0.00942873 | 0.00333170 | -2.8300 | 0.004655 **   |
| grade  | 0.00929152  | 0.00682659 | 1.3611  | 0.173489      |
| south  | 0.02174921  | 0.01529361 | 1.4221  | 0.154994      |
| tenure | -0.00083676 | 0.02524139 | -0.0332 | 0.973555      |
| gender | 0.08348932  | 0.01529146 | 5.4599  | 4.765e-08 *** |
| income | -0.00058492 | 0.00018957 | -3.0855 | 0.002032 **   |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:

[1] "south" "gender"

<sup>1</sup>The data set is available at <https://www.stata-press.com/data/r17/chicken.dta>.

<sup>2</sup>The exercise is based on Stata Reference Manual (xtprobit function).

```
> null_probit = glm(complain~1, data=chicken, family=binomial(link="probit"))
> lrtest(probit1, null_probit)
Likelihood ratio test
```

Model 1: complain ~ age + grade + south + tenure + gender + income

Model 2: complain ~ 1

```
#Df  LogLik Df Chisq Pr(>Chisq)
1    7 -1376.3
2    1 -1402.3 -6 52.11  1.773e-09 ***
```

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
> gof.results = gof(probit1)
```

```
> gof.results$gof
      test  stat      val df      pVal
1:      HL chiSq 4.9805893  8 0.7596487
2:      mHL      F 0.7304362  9 0.6813310
3:      OsRo      Z 1.2660267 NA 0.2055035
4: SstPgeq0.5      Z      NaN NA      NaN
5:  SstPl0.5      Z 0.3283324 NA 0.7426603
6:  SstBoth chiSq      NaN  2      NaN
7: SllPgeq0.5 chiSq 0.0000000  1 1.0000000
8:  SllPl0.5 chiSq 0.1081492  1 0.7422611
9:  SllBoth chiSq 0.1081492  2 0.9473614
```

Use  $\alpha = 5\%$  significance level.

- (5%) Interpret parameters' estimates.
- (5%) Interpret marginal effects.
- (5%) Are all variables jointly statistically significant? Explain.
- (7%) Let's focus on variable **tenure**. Is it endogenous? Explain.
- (8%) Interpret the diagnostic tests' results.







**Exercise 2 (30%)**

University administrators want to know the relationship between high school grade point average (GPA) and students' performance in college. `gpa.dta` contains fictional data on a cohort of 4,000 college students.<sup>3</sup> College GPA (`gpa2`) and high school GPA (`hsgpa`) are measured on a continuous scale between zero and four. The outcome of interest is the student's college GPA. But, for reasons of confidentiality, GPAs below 2.0 are reported as 2.0. In other words, the outcome is censored on the left.

We believe that GPA is also a function of the logarithm of income of the student's parents (`pincome`) and whether or not the student participated in a study-skills program while in college (`program`).<sup>4</sup>

The output below presents a tobit model estimates and the table presents marginal effects for this model.

Call:

```
censReg(formula = gpa2 ~ hsgpa + pincome + program, left = 2,
        right = Inf, data = gpa)
```

Observations:

| Total | Left-censored | Uncensored | Right-censored |
|-------|---------------|------------|----------------|
| 4000  | 1206          | 2794       | 0              |

Coefficients:

|             | Estimate  | Std. error | t value | Pr(>  t )  |
|-------------|-----------|------------|---------|------------|
| (Intercept) | -0.890258 | 0.047848   | -18.61  | <2e-16 *** |
| hsgpa       | 0.658631  | 0.012870   | 51.18   | <2e-16 *** |
| pincome     | 0.315930  | 0.007457   | 42.37   | <2e-16 *** |
| program     | 0.555442  | 0.014747   | 37.66   | <2e-16 *** |
| logSigma    | -0.910997 | 0.013606   | -66.95  | <2e-16 *** |

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 Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Newton-Raphson maximisation, 7 iterations

Return code 1: gradient close to zero

Log-likelihood: -2015.126 on 5 Df

```
> x = rbind(1, mean(gpa$hsgpa), mean(gpa$pincome), mean(gpa$program))
> me = tobit_marginal_effects(tobit1, x, dummies_indices=c())
```

Marginal effects of the tobit model

|         | y*        | E(y x)    | E(y x,y>0) | Pr(y>0 x)    | at X=    |
|---------|-----------|-----------|------------|--------------|----------|
| hsgpa   | 0.6586311 | 0.6586311 | 0.6586311  | 8.750838e-09 | 2.842114 |
| pincome | 0.3159297 | 0.3159297 | 0.3159297  | 4.197569e-09 | 3.680588 |
| program | 0.5554416 | 0.5554416 | 0.5554416  | 7.379821e-09 | 0.498500 |

<sup>3</sup>The data set is available at <https://www.stata-press.com/data/r16/gpa.dta>.

<sup>4</sup>The exercise is based on Stata Reference Manual (`tobit` function).

Use  $\alpha = 5\%$  significance level.

- a) (5%) Why shouldn't we use simple regression model for the problem? Explain.
- b) (7%) Is this a corner solution or censoring variable case? Explain.
- c) (5%) What is the interval of values of the dependent variable? How many observations did take values on the edge of the interval?
- d) (5%) Which variables are individually significant? Explain.
- e) (8%) What is the effect of participation in a study-skills program while in college? Use appropriate marginal effect.







**Exercise 3 (40%)**

One way of modeling supply response for an agricultural crop is to specify a model in which area planted (acres) depends on price. When the price of the crop's output is high, farmers plant more of that crop than when its price is low. Letting AREA denote area planted, and PRICE denote output price, and assuming a log-log (constant elasticity) functional form, a finite distributed lag area response model of this type can be written as

$$\ln AREA_t = \beta_0 + \sum_{i=1}^q \beta_i \ln PRICE_t + \sum_{i=1}^p \alpha_i \ln AREA_t + \varepsilon_t \quad (1)$$

We use this model to explain the area of sugar cane planted in a region of the southeast Asian country of Bangladesh. Information on the delay and interim elasticities is useful for government planning. It is important to know whether existing sugar processing mills are likely to be able to handle predicted output, whether there is likely to be excess milling capacity, and whether a pricing policy linking production, processing, and consumption is desirable. Data comprising 34 annual observations on area and price are given in the file `bangla.dat`.<sup>5</sup>

Use  $\alpha = 5\%$  significance level.

Time series regression with "zoo" data:

Start = 3, End = 34

Call:

```
dynlm(formula = lnAREA ~ L(lnAREA, c(1:2)) + lnPRICE + L(lnPRICE, c(1:2)),
      data = bangla)
```

Coefficients:

|                     | Estimate | Std. Error | t value | Pr(> t ) |    |
|---------------------|----------|------------|---------|----------|----|
| (Intercept)         | 2.40315  | 0.82830    | 2.901   | 0.00747  | ** |
| L(lnAREA, c(1:2))1  | 0.46907  | 0.19354    | 2.424   | 0.02263  | *  |
| L(lnAREA, c(1:2))2  | -0.08084 | 0.18965    | -0.426  | 0.67342  |    |
| lnPRICE             | 0.78292  | 0.29717    | 2.635   | 0.01401  | *  |
| L(lnPRICE, c(1:2))1 | -0.57225 | 0.33433    | -1.712  | 0.09887  | .  |
| L(lnPRICE, c(1:2))2 | 0.09622  | 0.33891    | 0.284   | 0.77872  |    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.295 on 26 degrees of freedom

Multiple R-squared: 0.3213, Adjusted R-squared: 0.1908

F-statistic: 2.462 on 5 and 26 DF, p-value: 0.05925

- (10%) Write down the empirical model. Is a DL or an ARDL model? Explain.
- (10%) What is the steady-state solution for  $\ln PRICE^* = 1$ ?
- (10%) Calculate and interpret the long term multiplier.
- (5%) What are variables the model could be extended with? Explain your proposal.
- (5%) What do we have to check whenever such model is estimated?

<sup>5</sup>This exercise is based on Exercise 9.14 from Guay C. Lim, R. Carter Hill, and William Griffiths (2008). *Principles of Econometrics*, Wiley.







